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CIA - 1

① Given,

metropolitan region = 1200 km²

Radius = 1.8 km

spectrum = 48 MHz

full duplex = 200 KHz

N = 7 & n = 4

Minimum reuse ratio = 4:5

Step 1:

Area of hexagonal cell

$$A_c = \frac{3\sqrt{3}}{2} R^2$$

R = 1.8

$$A_c = \frac{3\sqrt{3}}{2} (1.8)^2 = 2.598 (3.24)$$

$$A_c = 8.4175 \text{ km}^2$$

No of cells = $N_c = \frac{1200}{8.4175}$

$$N_c = 143$$

Total Available channels.

$$C = \frac{\text{Total spectrum}}{\text{Band width}}$$

$$C = \frac{48 \times 1000}{200}$$

C = 240 channels

After 15% reduction

Effective = 240 × 0.85

= 204 channels

Theoretical & effective channels per cell

N = 7

Theoretical channels per cell

$C_7 = \frac{240}{7} = 34.29$

Effective channels per cell

C_7 eff = 34.29

= 29.17

N = 4

Theoretical channels per cell

$C_4 = \frac{240}{4} = 60$

Effective channels per cell

C_4 eff = $\frac{204}{4}$ = 51

C_4 eff = 51

Co channel reuse distance

$D = R\sqrt{3N}$

N = 7

$n = 7$

$D = 1.8\sqrt{3(7)}$

$= 1.8\sqrt{21}$

$= 8.25$ km

$n = 4$

$D = 1.8\sqrt{3(4)}$

$= 1.8\sqrt{12}$

$= 6.24$ km

(2) Given,

Planned emergency operation duration = 40 mins

No of cell boundaries = 15

Speed = 40 km/h

B_s = 120 ms

n_{bs} = 80 ms

$T = T_{BS} + T_{NB}$

$= 120 + 80$

(3) $C = \frac{\text{Total spectrum}}{CR} = \frac{1000}{20}$

Channels per cell

$$C_{cell} = \frac{333}{7} = 47$$

12 Base transceiver stations

3 sectors and 4 transceivers (TRX)

It serves 1200 subscribers

TDMA Total channel

No of cells

$$c = \frac{10000}{200} = 100$$

Carrier - 8 - use

Total cell

$$50 \times 8 = 400$$

Channel per cell.

$$C_{cell} = \frac{400}{4} = 100$$

$$P_G = \frac{\text{Chip rate}}{\text{Data rate}} = \frac{1221}{9600}$$

CDMA model

$$IL = \frac{P_G}{E_b/N_o} \times VF$$

$$= \frac{128}{8} \times 0.4 \times 0.1$$

Total CDMA = $C = \frac{40}{1.25} \cdot 8$

$= 8 \times 3.84$

$= 30.72$

$$\text{Gain} = \frac{\text{CDMA} - \text{FDMA}}{\text{FDMA}} \times 100$$

Compand FDMA = $\frac{31.72 - 47.57}{47.37} \times 100 = 34.5$

TDMA = $\frac{30.72 - 100}{100} \times 100 = 69.28$

(4) Medical imaging file = 240

Imaging size = 18mb

Data rate = 20kbps

throughput = 85%

Data = No. of file

$= 240 \times 18$

4320×8

80%.

$R_1 = 80 \times 0.82$

$= 65.6$

85%.

$R_e = 80 \times 0.85$

$R_e = 68$

Data volume

$D_v = 432 \times 1.42$

$m_b = 4838.4 \times 8$

$= 38707.2$

mobility =

$T = 8 \times 2.2$

$= 18.4$

Frame rate

$T = \frac{bv}{theory}$

$= \frac{38707}{35.76} = 1.08$

(5) Given,

12 GB software update to 2000 devices

GSM = 9.6 KBPS

throughput = 64 KBPS with 10% additional transmission traffic

5G =

25 MBPS per device

4% protocol overhead

1 GB = 1000 MB

1 MB = 8 mega bits

=> 1000 x 8

= 8000 mega bits = 1 GB

① 120 x 1000 (Data volume per device)

= 120 000 MB

=> $\frac{120000}{2000}$

=> 60 MB = 480 mega bits

② $\frac{480 \times 1000}{9.6}$ (GSM transfer time per device)

= 50,000 S

= 13.89 hours

③ CPU transfer time

$$= \frac{48\text{mb} \times 1.10 \times 1000}{64}$$

$$= 825\text{s} = 2.29\text{hours}$$

④ Effective SU throughput

$$\Rightarrow 25\text{V}(1-0.04)$$

$$= 24\text{ mbps}$$

⑤ SU transfer per device

$$\Rightarrow \frac{480}{24} = 20.0\text{S}$$